

# Optimization

SGD

B - batch size

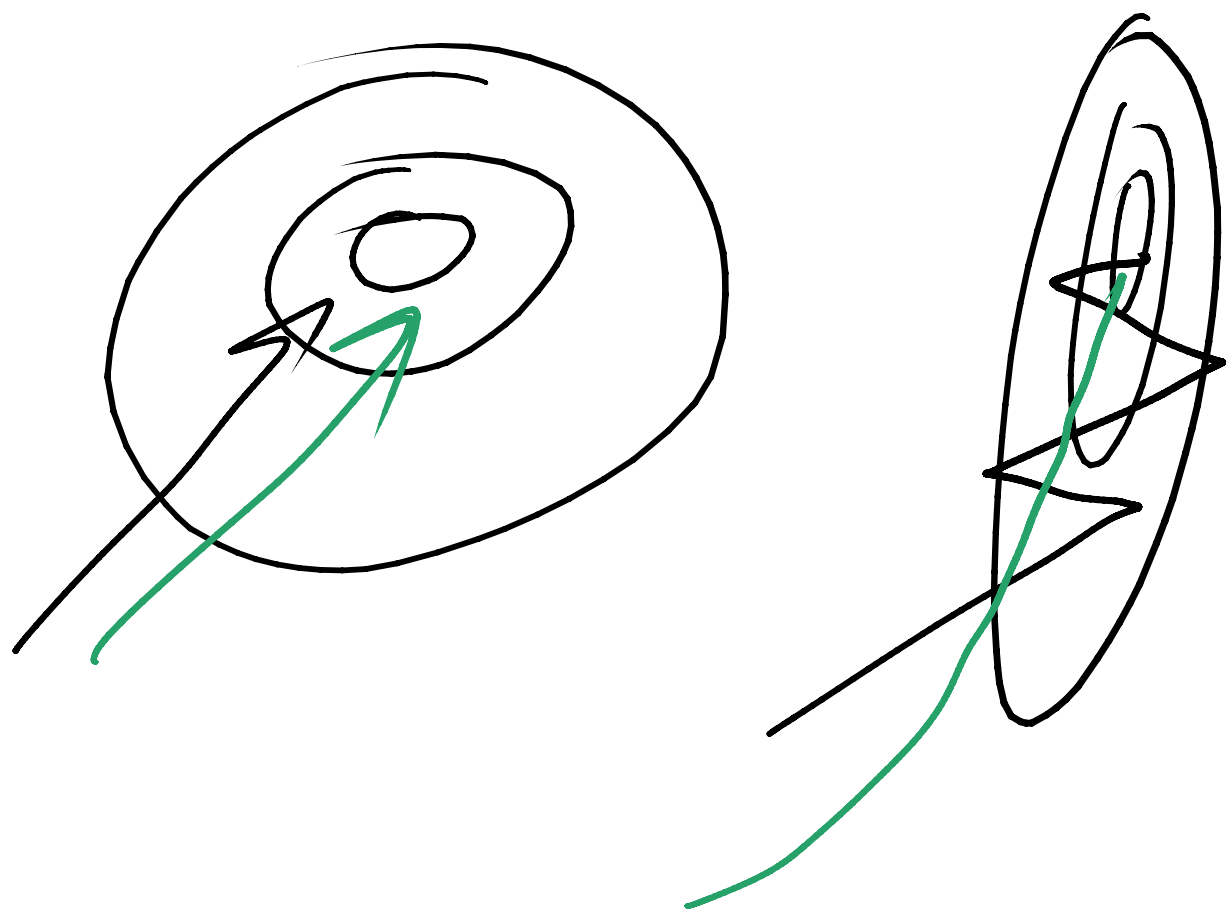
$$\mathcal{L}^t(w) = \frac{1}{B} \sum_{i=1}^B \mathcal{L}(f_w(x_{t_i}), y_{t_i})$$

$$t_i \sim U\{1, \dots, \ell\} \quad \ell - \text{train size}$$

$$\begin{cases} g_t = \nabla_w \mathcal{L}^{t-1}(w_{t-1}) \\ m_t = \mu m_{t-1} + g_t \\ w_t = w_{t-1} - \eta m_t \end{cases} \quad \text{SGD} + \text{momentum}$$

$$m_0 = 0$$

$$0 \leq \mu < 1 \quad \approx 0.9$$



$$\mathcal{L}^t(w) \rightarrow \mathcal{L}_\lambda^t(w) = \mathcal{L}^t(w) + \frac{\lambda}{2} \|w\|_2^2$$

$$\nabla_w \|w\|_2^2 = 2w$$

$$\begin{aligned}
 & \begin{cases} g_t = \nabla \omega \alpha^{t-1} (w_{t-1}) + \lambda w_{t-1} \\ m_t = \mu m_{t-1} + g_t \\ w_t = w_{t-1} - \eta m_t = \end{cases} \\
 & = w_{t-1} - \eta (\mu m_{t-1} + g_t) = \\
 & = w_{t-1} - \eta (\mu m_{t-1} + \nabla \omega \alpha^{t-1} (w_{t-1}) + \lambda w_{t-1}) = \\
 & = w_{t-1} (1 - \eta \lambda) - \eta (\mu m_{t-1} + \\
 & + \nabla \omega \alpha^{t-1} (w_{t-1}))
 \end{aligned}$$

weight decay

$$\lambda \approx 10^{-4}, 10^{-5}$$

Adam

- AdaGrad / RMS Prop
- Momentum

$$\begin{cases} m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t \\ v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t \odot g_t \end{cases}$$

$\beta_1, \beta_2$  — hyperparams  $\beta_1 = 0.9$   $\beta_2 = 0.999$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1-\beta_2^t} \quad \begin{matrix} m_0 = \vec{0} \\ v_0 = \vec{1} \end{matrix}$$

$$w_t = w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} \quad \eta_t - \text{learning rate}$$

$$\epsilon \approx 10^{-8}$$

$$\begin{aligned} m_t &= (1-\beta_1) g_t + \beta_1 m_{t-1} = \\ &= (1-\beta_1) g_t + \beta_1 (1-\beta_1) g_{t-1} + \beta_1^2 m_{t-2} = \\ &= \sum_{k=0}^{t-1} (1-\beta_1) \beta_1^k g_{t-k} \end{aligned}$$

$$\hat{m}_t = \frac{1}{1-\beta_1^t} \quad m_t = \frac{1-\beta_1}{1-\beta_1^t} \sum_{k=0}^{t-1} \beta_1^k g_{t-k} =$$

$$= \sum_{k=0}^{t-1} \beta_1^k g_{t-k}$$

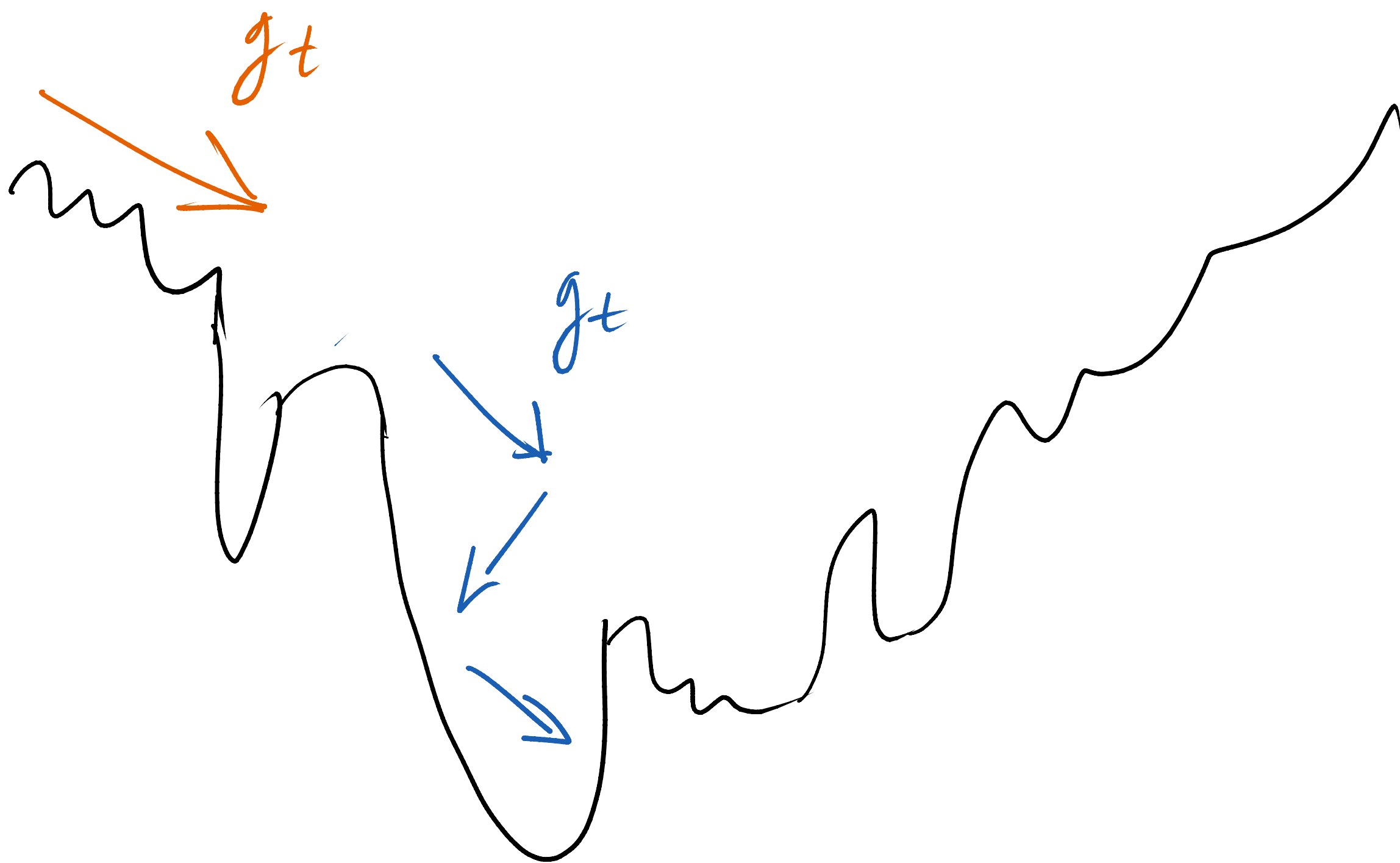
$$\frac{\sum_{m=0}^{t-1} \beta_1^m}{\sum_{m=0}^{t-1} \beta_1^m} \quad \text{⊖}$$

$$\alpha_k = \frac{\beta_1^k}{\sum_{m=0}^{t-1} \beta_1^m}$$

$$\sum \alpha_k = 1$$

$$\text{⊖} \quad \sum_{k=0}^{t-1} \alpha_k g_{t-k} \simeq \mathbb{E}g$$

$$\hat{v}_t \simeq \mathbb{E}[g \odot g] = \text{Var}(g) + \mathbb{E}[g] \cdot \mathbb{E}[g]$$



1. Far from optimum

$$\text{Var}(g) \approx 0$$

$$\eta_t \frac{\hat{m}_t}{\sqrt{\hat{V}_t + \epsilon}} \approx \frac{\mathbb{E}g}{\sqrt{\cancel{\text{Var}(g)} + \mathbb{E}g \circ \mathbb{E}g + \epsilon}} \approx 1$$

$\swarrow$   
 $0$

2. Close to optimum

$$\text{Var}(g) \gg 0 \quad \frac{\mathbb{E}g}{\sqrt{\text{Var}(g) + \mathbb{E}g \circ \mathbb{E}g + \epsilon}} < 1$$

SGD + momentum

convolutional

Adam

RNN, Transformers

$$w_t = w_{t-1} - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$

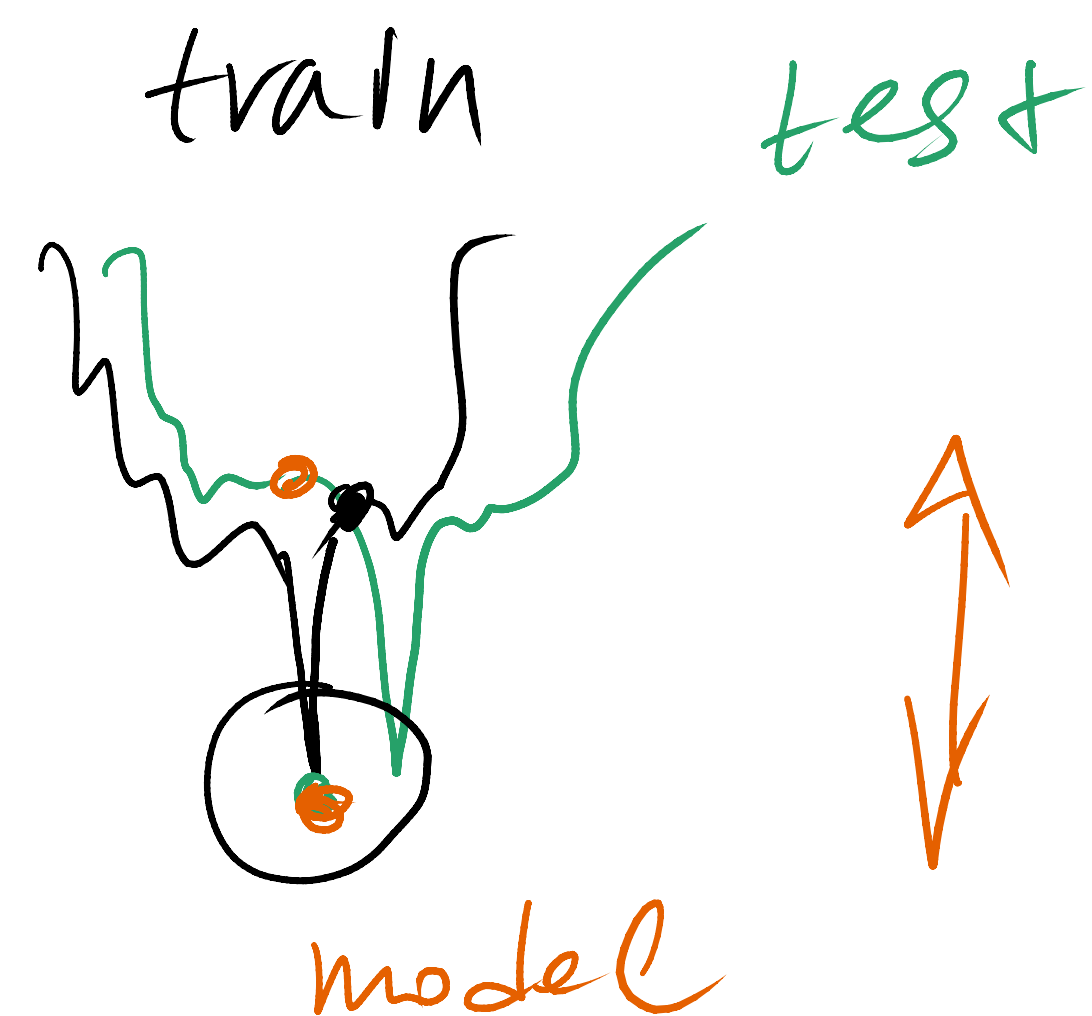
$$= w_{t-1} - \eta_t \frac{m_t}{(1 - \beta_1^t) \sqrt{\hat{v}_t + \epsilon}}$$

$$= w_{t-1} - \eta_t \frac{\beta_1 m_{t-1} + (1 - \beta_1) \nabla_w \mathcal{L}^{t-1}(w_{t-1}) + \lambda w_{t-1}}{(1 - \beta_1^t) \sqrt{\hat{v}_t + \epsilon}}$$

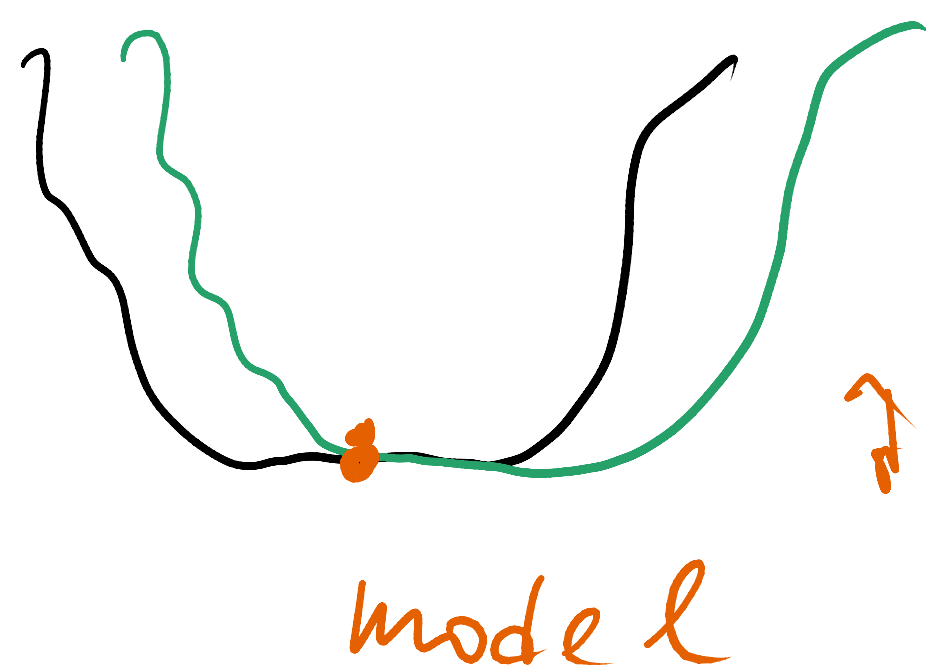
$$= w_{t-1} \left( 1 - \frac{\eta_t \lambda}{(1 - \beta_1^t) \sqrt{\hat{v}_t + \epsilon}} \right) - \dots$$

AdamW — Transformers

$$= w_{t-1} (1 - \eta_t \lambda) - \eta_t \frac{\beta_1 w_{t-1} + (1 - \beta_1) \nabla_w \mathcal{L}^{t-1}(w_{t-1})}{(1 - \beta_1^t) \sqrt{\hat{v}_t + \epsilon}}$$



sharp minimum



flat minimum

SAM Sharpness Aware Minimization  
 $\forall p > 0$  with high proba

$$\mathcal{L}_{\text{test}}(w) \leq \max_{\|\epsilon\|_2 \leq p} \mathcal{L}_{\text{train}}(w + \epsilon) +$$

$$+ h(\|w\|_2^2 / p^2) \quad h: \mathbb{R}^+ \rightarrow \mathbb{R}^+ \uparrow$$

$$\max_{\|\varepsilon\|_2 \leq \rho} \left( \mathcal{L}_{\text{train}}(w + \varepsilon) - \mathcal{L}_{\text{train}}(w) \right) +$$

$$+ \mathcal{L}_{\text{train}}(w) \times h(\dots)$$

$$\max_{\|\varepsilon\|_2 \leq \rho} \left( \mathcal{L}_{\text{train}}(w) + \varepsilon^T \nabla_w \mathcal{L}_{\text{train}}(w) \right)$$

Wide nets

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SWA

$$\hat{w}_t = \mu \hat{w}_{t-1} + (1-\mu) w_t$$

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